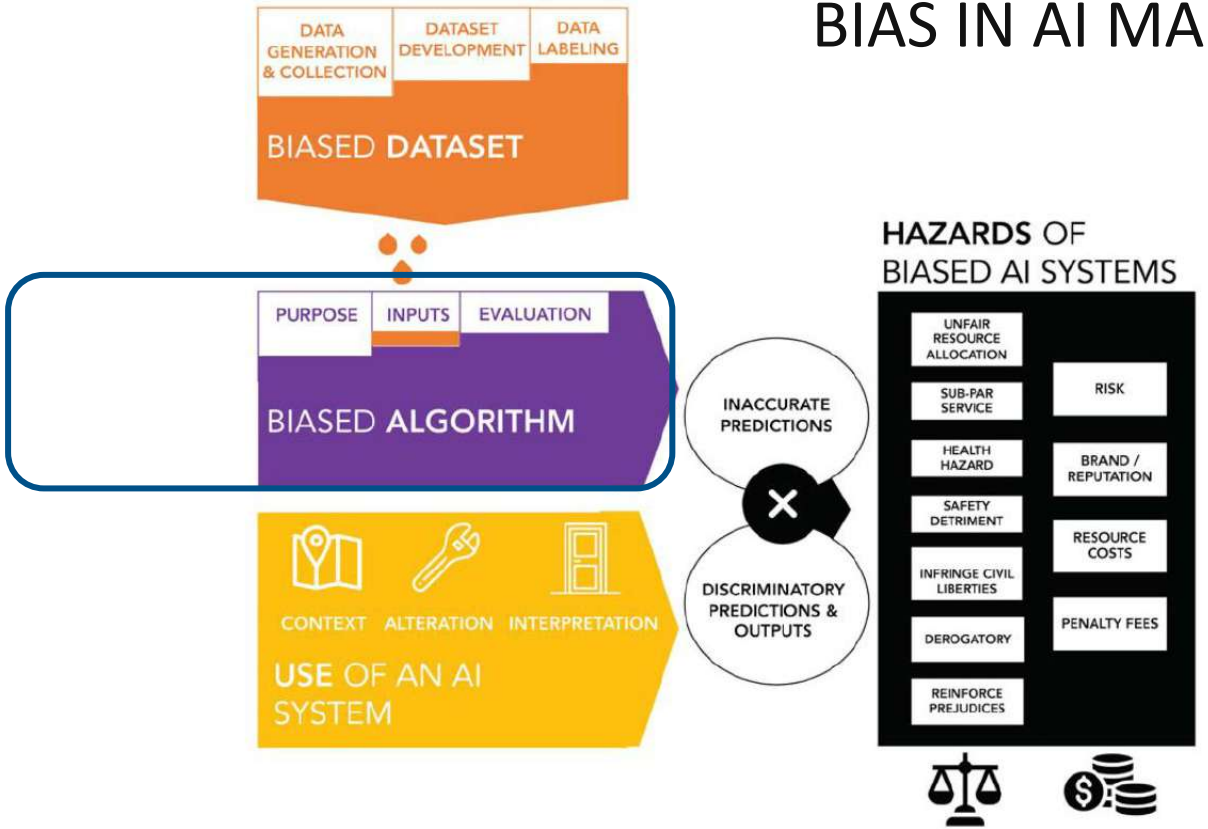
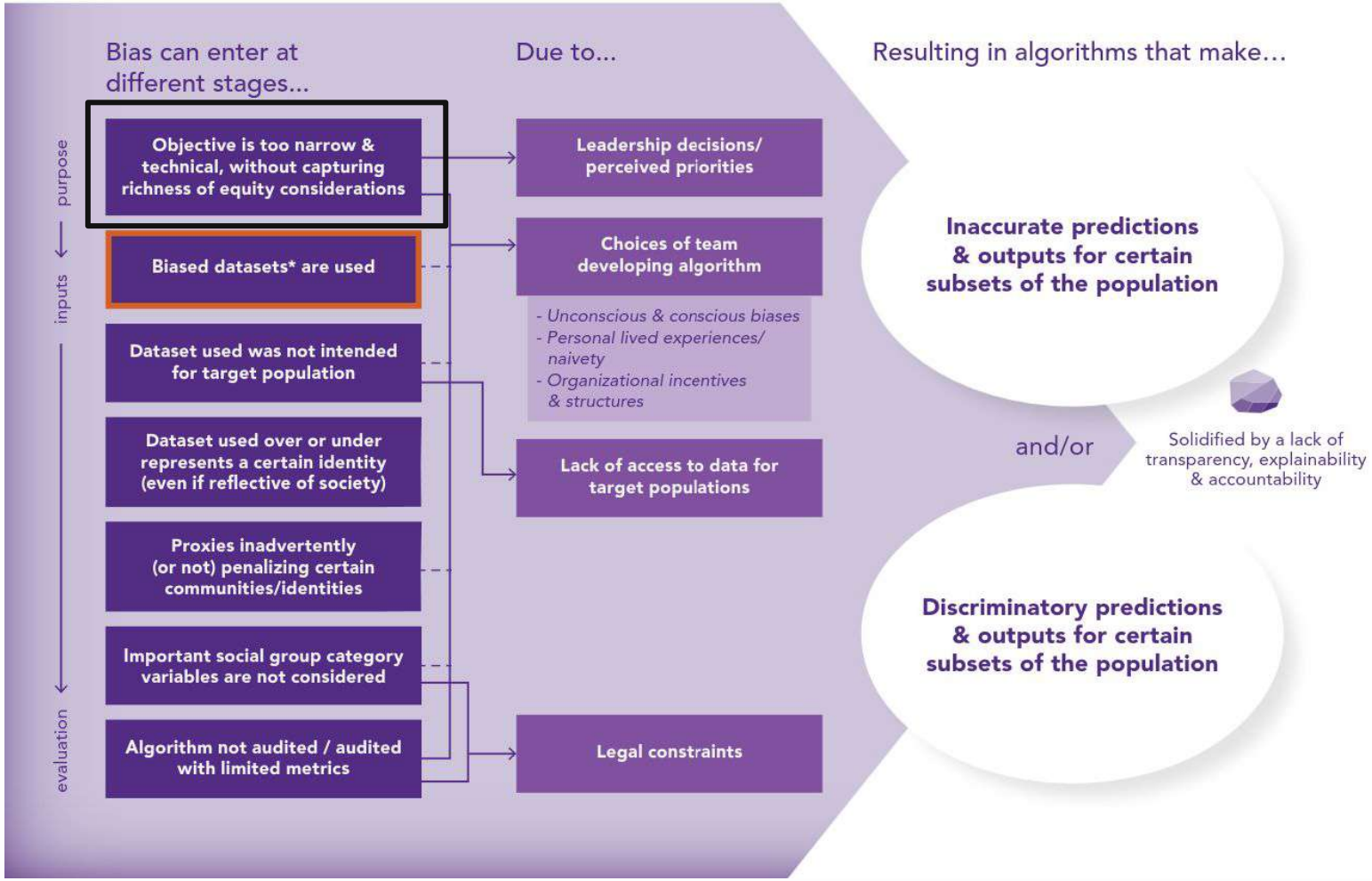
Why are AI systems biased?

BIAS IN AI MAP



Mitigating Bias in Artificial Intelligence, Haas School of Business, University of California, Berkeley


Bias can enter at different stages...
Due to...
Resulting in algorithms that make...
purpose
inputs
evaluation
Objective is too narrow & technical, without capturing richness of equity considerations
Biased datasets* are used
Dataset used was not intended for target population
Dataset used over or under represents a certain identity (even if reflective of society)
Proxies inadvertently (or not) penalizing certain communities/identities
Important social group category variables are not considered
Algorithm not audited / audited with limited metrics
Leadership decisions/ perceived priorities
Choices of team developing algorithm
- Unconscious & conscious biases
- Personal lived experiences/ naivety
- Organizational incentives & structures
Lack of access to data for target populations
Legal constraints
Inaccurate predictions & outputs for certain subsets of the population
and/or
Solidified by a lack of transparency, explainability & accountability
Discriminatory predictions & outputs for certain subsets of the population

Algorithm purpose

SUPREME COURT

June 2023

Supreme Court strikes down college affirmative action programs

The ruling is the culmination of decades of effort to end the consideration of race in admissions.

Running an ML algorithm with the **sole objective of minimizing error rates (efficiency)** (as is done typically), can mean missing other mistakes like **how error rates differ for different demographics (fairness).**

AEA Papers and Proceedings 2018, 108: 22–27
https://doi.org/10.1257/pandp.20181018

ADVANCES IN BIG DATA RESEARCH IN ECONOMICS

Algorithmic Fairness[1]

By Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Ashesh Rambachan*

Ending Affirmative Action Harms Diversity Without Improving Academic Merit

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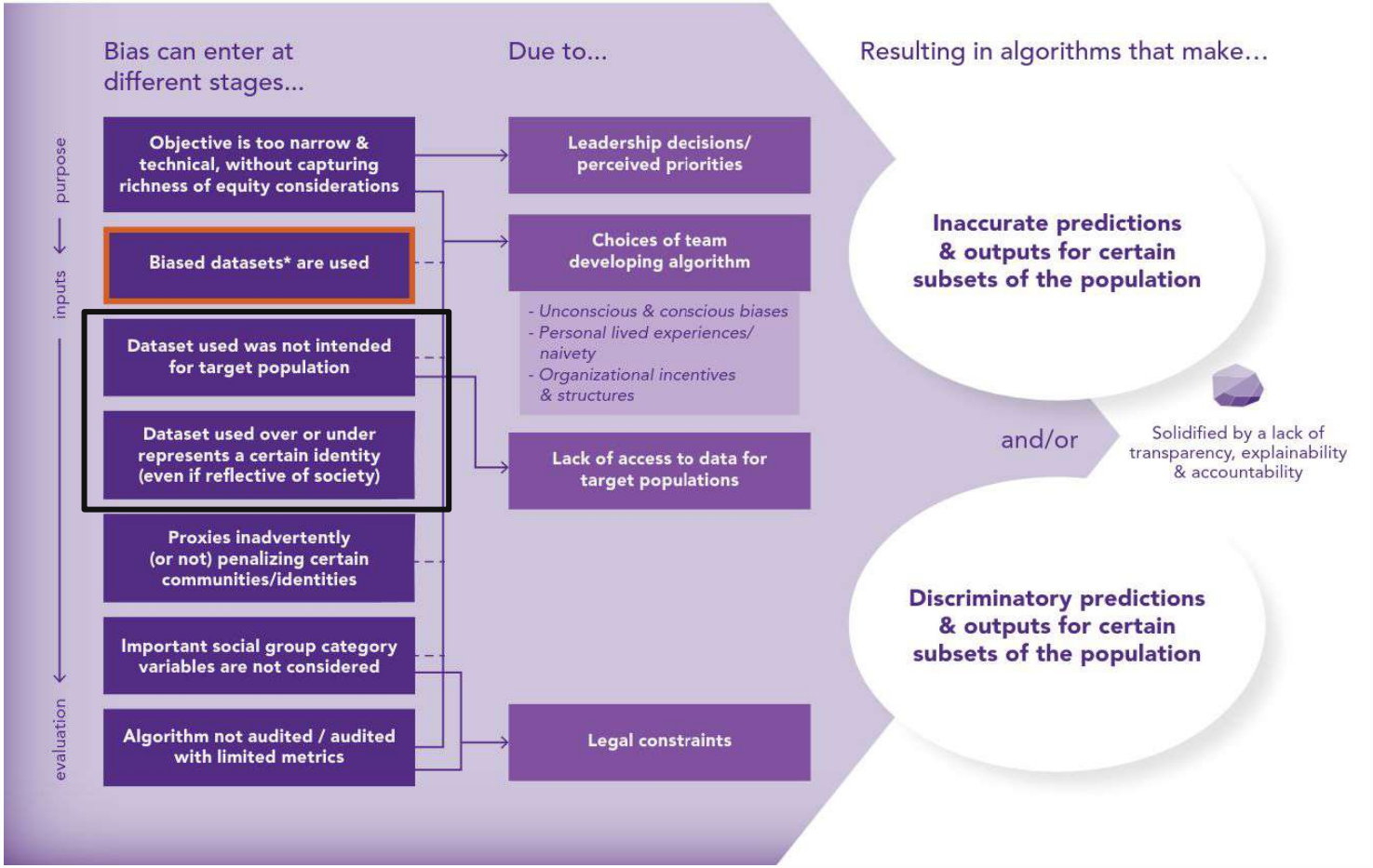
EAAMO '24, https://dl.acm.org/doi/pdf/10.1145/3689904.3694706

Researchers examined how college admission algorithms predicting GPA could be improved by **prioritizing equity over efficiency.**

Using an equity lens (considering race) rather than an efficiency lens (focusing only on predicted grades) **results in a fairer racial distribution** and **higher average grades among admitted students.**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What if we allow uses of a skin-care medicine in India, which is trialed and tested in Europe?

What if we have automated vehicles in India which are trained and tested on traffic dataset from other countries?

Algorithm inputs

- In 2018, Amazon recalled its experimental AI-driven hiring tool due to gender bias.
- **The tool penalized resumes containing the word “women’s.”**
- It was trained on a dataset of past applicants over 10 years, mostly men.
- The AI learned from historical gender disparities in the tech sector.
- **Instead of eliminating bias, it reinforced and exacerbated it.**

Reuters

World Business Markets Sustainability Legal Breakingviews Technology Investigatio

Insight - Amazon scraps secret AI recruiting tool that showed bias against women

By Jeffrey Dastin

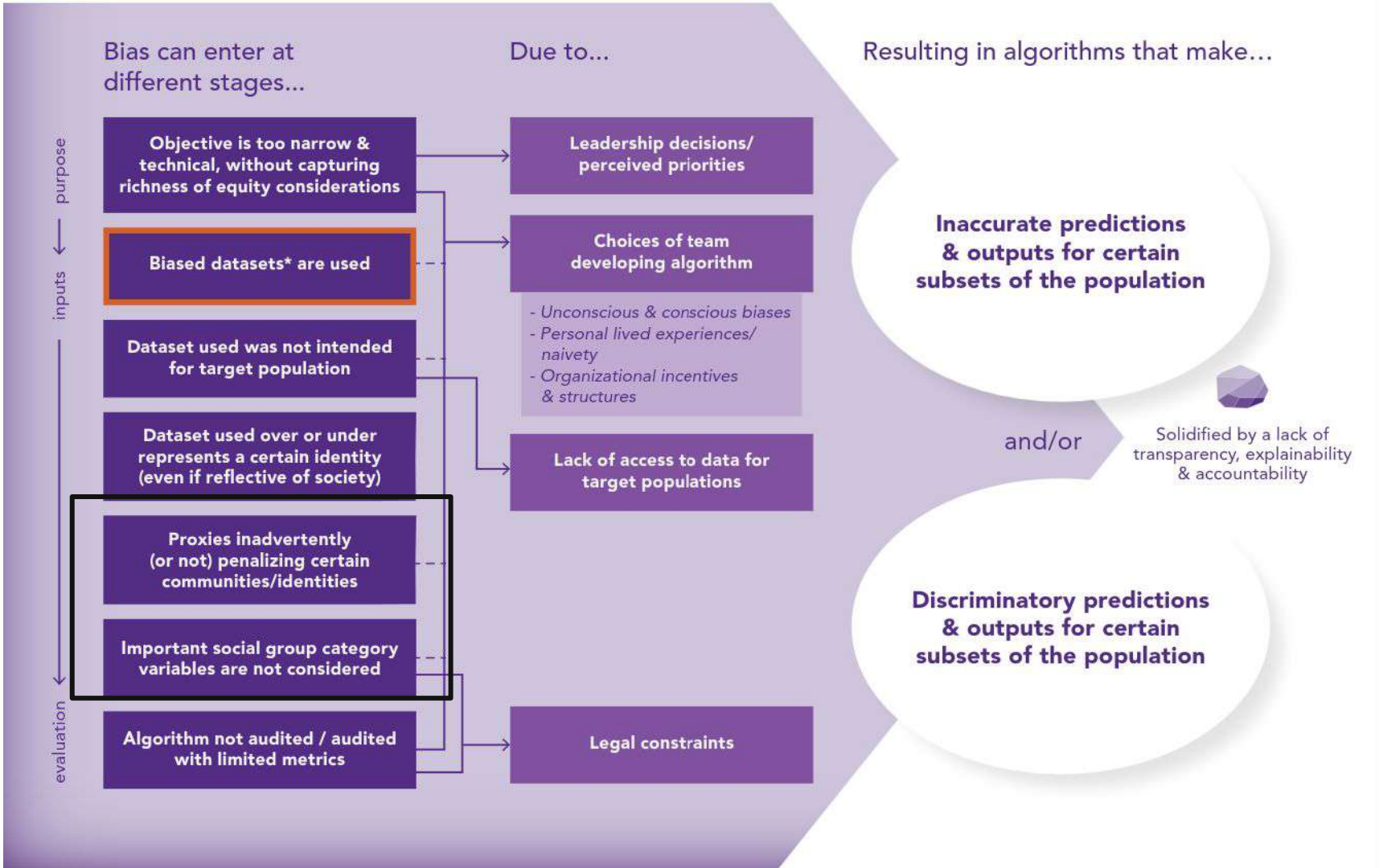
October 11, 2018 6:20 AM GMT+5:30 · Updated 6 years ago

Aa

SAN FRANCISCO (Reuters) - Amazon.com Inc's machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

The team had been building computer programs since 2014 to review job applicants' resumes with the aim of mechanizing the search for top talent, five people familiar with the effort told Reuters.

Automation has been key to Amazon's e-commerce dominance, be it inside warehouses or driving pricing decisions. The company's experimental hiring tool used artificial intelligence to give job candidates scores ranging from one to five stars - much like shoppers rate products on Amazon, some of the people said.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Proxies / variables

Proxies are defined as a variable used instead of the variable of interest when that variable of interest cannot be measured directly.

For example, per capita GDP (Gross domestic product) can be used as a proxy for the standard of living,
Body Mass Index (BMI) instead of True body fat.

How is “healthcare spending as a proxy for medical needs?”

- In one algorithm widely used in the US, researchers found it falsely concluded that **Black patients are healthier than equally sick White patients.**

- Used **health costs as a proxy for needs**, overlooking disparities in access to care.

- Black patients who **have the same level of need spend less money because of unequal access to care**, among other reasons.

- Reformulating **the algorithm to no longer use costs as a proxy for needs more than doubles the number of Black patients** receiving adequate care.

Dissecting racial bias in an algorithm used to manage the health of populations,
https://www.ftc.gov/system/files/documents/public_events/1548288/privacycon-2020-ziad_obermeyer.pdf

- Algorithm excluded race in their data, therefore the developers considered their algorithm to be “race-blind”.

There is no such thing as race in health-care algorithms,
https://www.thelancet.com/journals/landig/article/PIIS2589-7500(19)30201-8/fulltext#:~:text=The%20researchers%20found%20that%20the,in%20the%20USA%20each%20year.

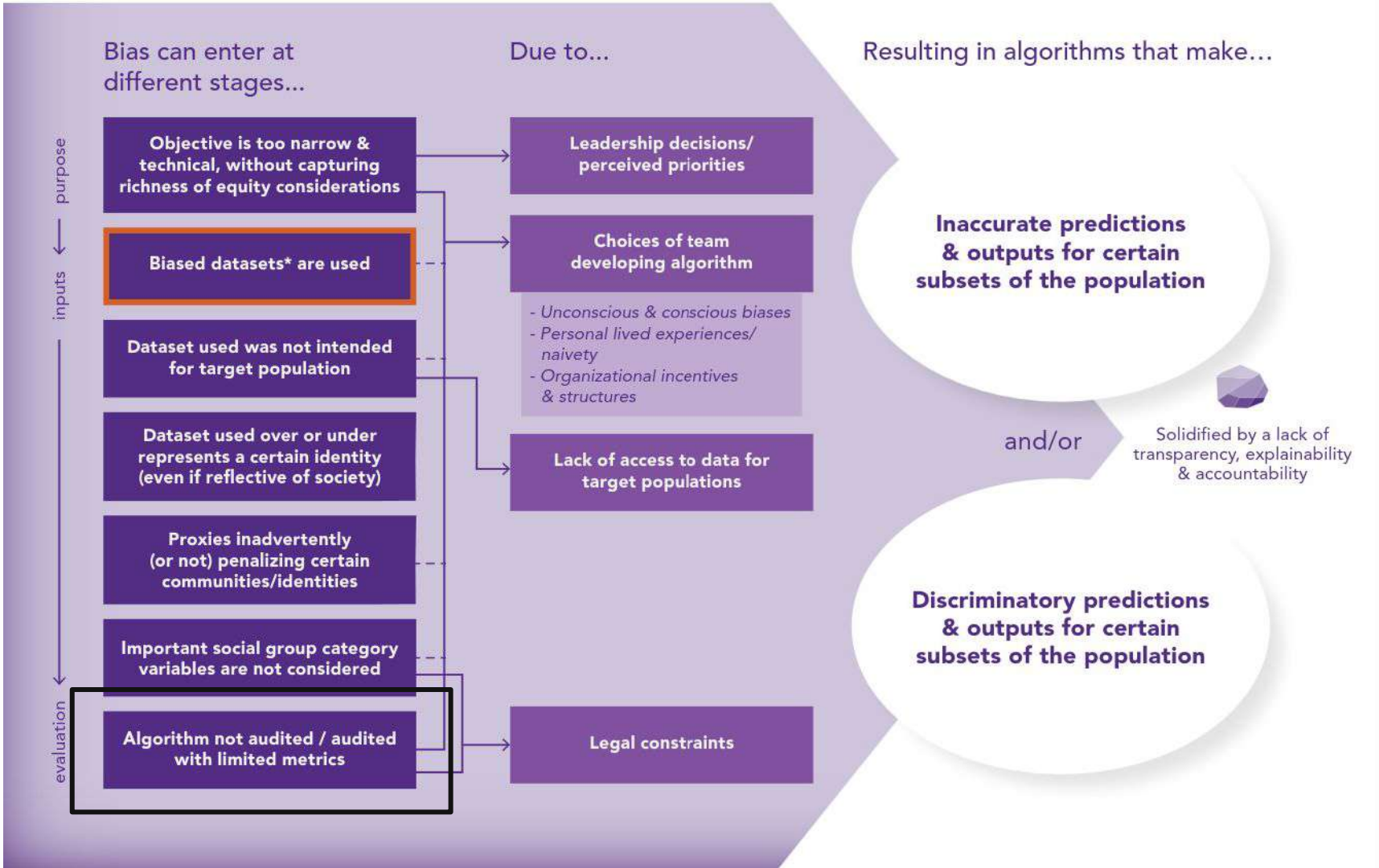
Proxies / variables

Early credit assessment used **discriminatory factors like clothing, gender, and race.**

These were later replaced with ***seemingly*** **neutral financial history data**.

But, past discrimination left women and people of socially underprivileged community with less formal financial history and discrimination, impacting their ability to get credit.

As a result, **AI systems today continue to reinforce inequitable access to credit along race and gender lines.**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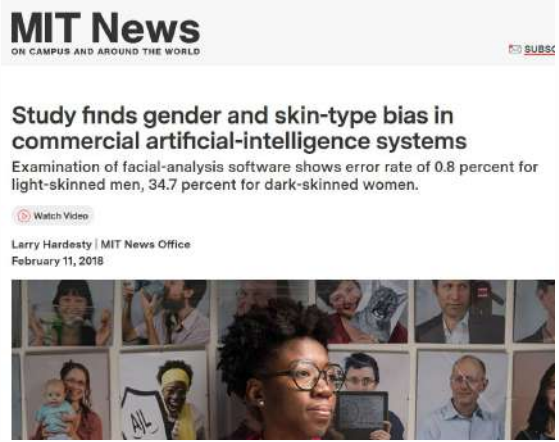
Algorithm evaluation

Auditing is crucial for identifying issues in an algorithm.

Limited or selective auditing can introduce harmful bias.

If a team focuses **only on accuracy and ignores false positives across demographics, errors may go unnoticed, leading to harm from untested conditions.**

COMPAS: Racial Bias in Criminal Justice Systems



MIT News

ON CAMPUS AND AROUND THE WORLD

Study finds gender and skin-type bias in commercial artificial-intelligence systems

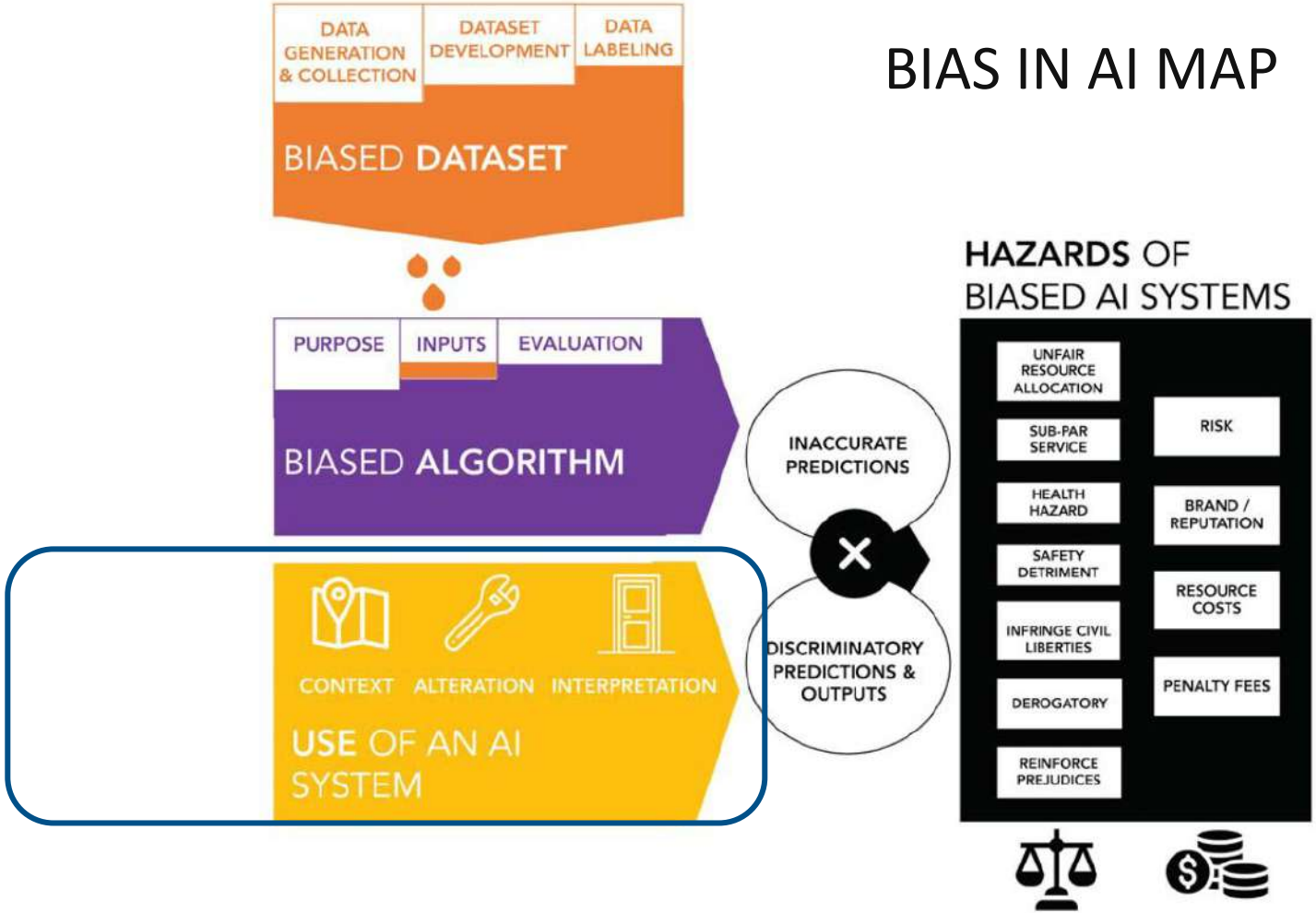
Examination of facial-analysis software shows error rate of 0.8 percent for light-skinned men, 34.7 percent for dark-skinned women.

Watch Video

Larry Hardesty | MIT News Office

February 11, 2018

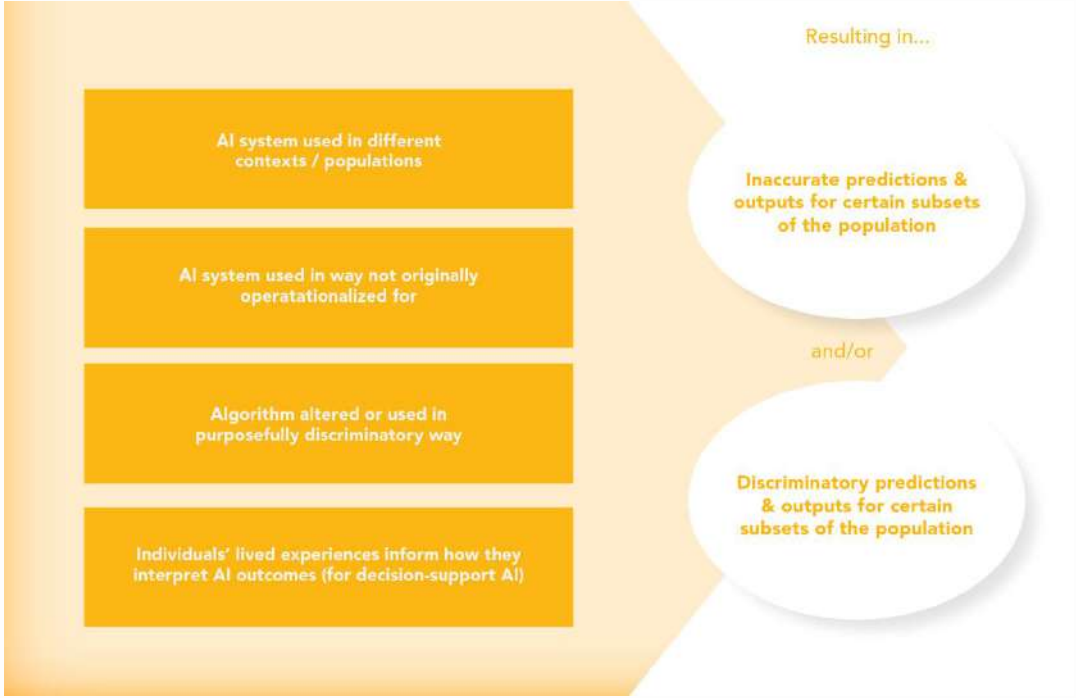
BIAS IN AI MAP



Mitigating Bias in Artificial Intelligence, Haas School of Business, University of California, Berkeley

USE OF AI SYSTEMS

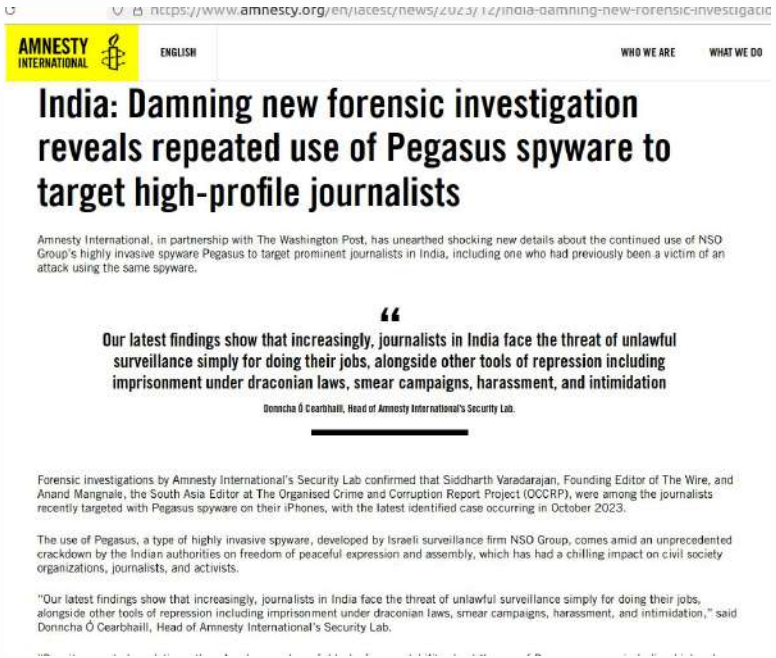
Beyond a biased algorithm(s), AI systems can result in discriminatory outcomes for certain individuals or populations based on how they are used.



USE OF AI SYSTEMS

Use of facial recognition technology for predictive policing.

In India, the **Pegasus** Project investigations alleged that the spyware was used to target ministers, opposition leaders, journalists, activists, minority leaders, Supreme Court judges, and key officials like CBI and EC.



https://www.amnesty.org/en/latest/news/2023/12/india-damning-new-forensic-investigatio

AMNESTY INTERNATIONAL | ENGLISH | WHO WE ARE | WHAT WE DO

India: Damning new forensic investigation reveals repeated use of Pegasus spyware to target high-profile journalists

Amnesty International, in partnership with The Washington Post, has unearthed shocking new details about the continued use of NSO Group's highly invasive spyware Pegasus to target prominent journalists in India, including one who had previously been a victim of an attack using the same spyware.

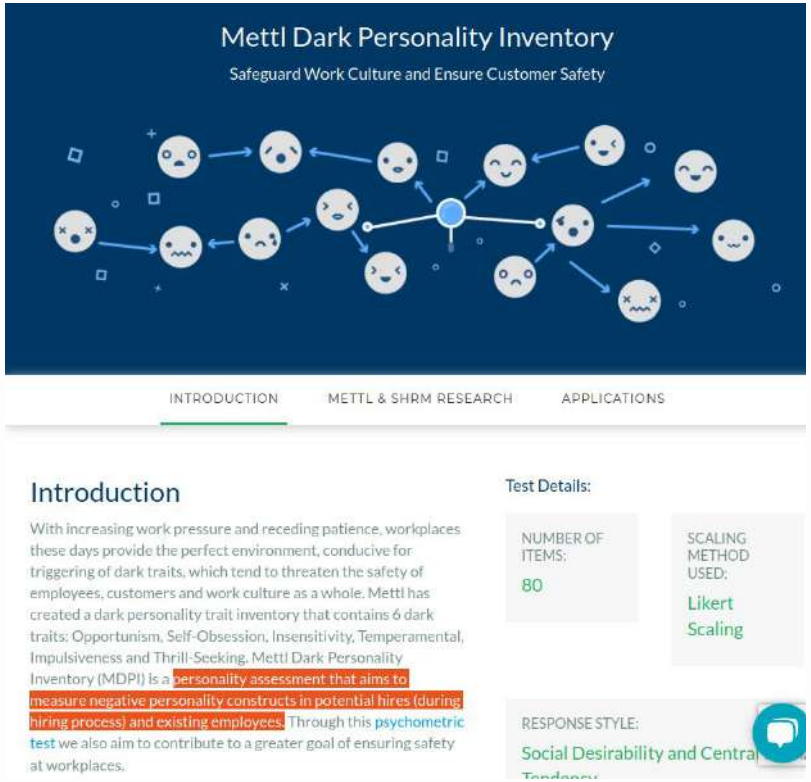
> **Our latest findings show that increasingly, journalists in India face the threat of unlawful surveillance simply for doing their jobs, alongside other tools of repression including imprisonment under draconian laws, smear campaigns, harassment, and intimidation**
>
> Donncha Ó Cearbhaill, Head of Amnesty International's Security Lab.

Forensic investigations by Amnesty International's Security Lab confirmed that Siddharth Varadarajan, Founding Editor of The Wire, and Anand Mangnale, the South Asia Editor at The Organised Crime and Corruption Report Project (OCCRP), were among the journalists recently targeted with Pegasus spyware on their iPhones, with the latest identified case occurring in October 2023.

The use of Pegasus, a type of highly invasive spyware, developed by Israeli surveillance firm NSO Group, comes amid an unprecedented crackdown by the Indian authorities on freedom of peaceful expression and assembly, which has had a chilling impact on civil society organizations, journalists, and activists.

"Our latest findings show that increasingly, journalists in India face the threat of unlawful surveillance simply for doing their jobs, alongside other tools of repression including imprisonment under draconian laws, smear campaigns, harassment, and intimidation," said Donncha Ó Cearbhaill, Head of Amnesty International's Security Lab.

USE OF AI SYSTEMS



Mettl has created a dark personality trait inventory that contains 6 dark traits: Opportunism, Self-Obsession, Insensitivity, Temperamental, Impulsiveness and Thrill-Seeking. Mettl Dark Personality Inventory (MDPI) is a **personality assessment that aims to measure negative personality constructs in potential hires (during hiring process) and existing employees.**

USE OF AI SYSTEMS

CENTER FOR THE ADVANCED STUDY OF INDIA
CASI
UNIVERSITY OF PENNSYLVANIA

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Mettl Dark Personality Inventory

Safeguard Work Culture and Ensure Customer Safety

INTRODUCTION ME

Introduction

With increasing work pressure and receding patie
these days provide the perfect environment, cond
triggering of dark traits, which tend to threaten th
employees, customers and work culture as a whol
created a dark personality trait inventory that co
traits: Opportunism, Self-Obsession, Insensitivity
Impulsiveness and Thrill-Seeking. Mettl Dark Per
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hiring process) and existing employees. Through t
test we also aim to contribute to a greater goal of
at workplaces.

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News › Company › Corporate Trends › Tech Mahindra adopts facial recognition to mark attendance

ETPrime

Tech Mahindra adopts facial recognition to mark attendance

Singh, ET Bureau | Last Updated: Aug 07, 2018, 02:58:00 PM IST

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Synopsis
Staff at TechM can now mark their attendance by simply looking into the new facial-recognition terminal.

Tech Mahindra chief people officer Harshvendra

NEW DELHI: Facial recognition could soon jump from your smartphone to your workplace with employers using it to mark attendance and gauge the mood of the workforce.

India in Transition

Not Just Privacy: The Dangers of Emotion Recognition Technology

BY VIDUSHI MARDA
August 14, 2023
Politics
Science & Technology
Society & Culture
सिर्फ़ निजता का सवाल नहीं है: भावनाओं को पहचानने वाली टैक्नोलॉजी के खतरे

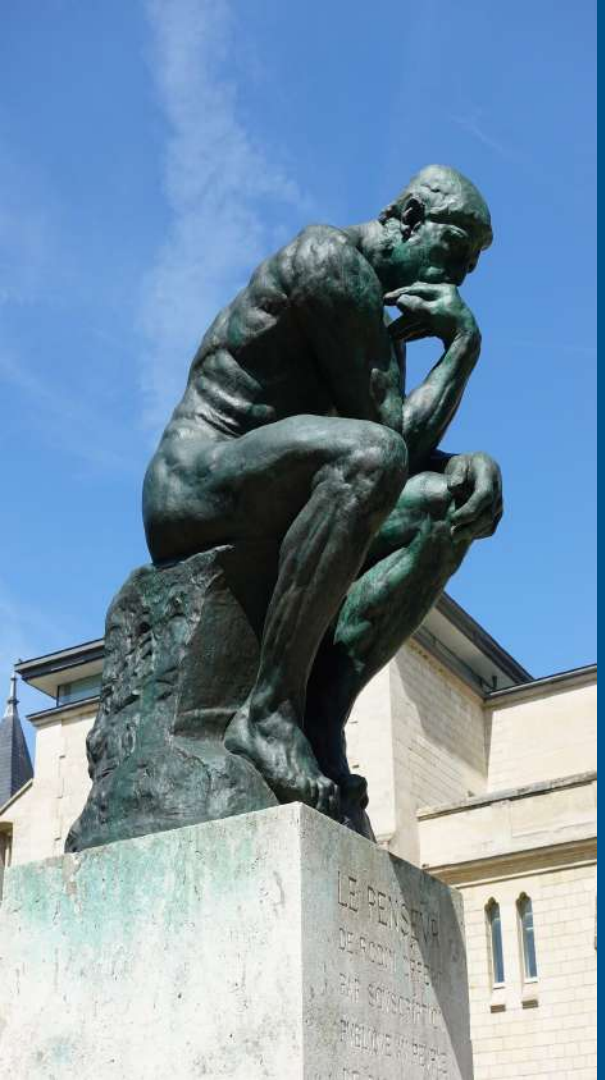
India's market for emotion recognition technologies has grown significantly in the last few years. AI start-ups are offering Indian employers the "dark personality inventory" that claims to identify "negative" traits like self-

USE OF AI SYSTEMS

- If an emotion recognition system flags you as temperamental and thrill-seeking, aka, a person with multiple “dark traits,” it can be difficult or impossible to prove or disprove this assumption.

- In 2021, the Uttar Pradesh police put forth a tender for its Safe City Initiative that included a requirement for the winning bid to include AI systems that would be able to detect “distressed” women.

- Sanitation workers in Haryana have been provided with smart watches equipped with microphones and GPS trackers so that their movements are seen and heard by supervisors.



Question 1: An online tech hiring platform, Gild, enables employers to use 'social data' (in addition to other resources such as resumes) to rank candidates. 'Social data' is a proxy that refers to how integral a programmer is to the digital community, drawing from time spent sharing and developing code on development platforms such as GitHub.

Is using such "social data" an unbiased strategy?

If it is biased, write with an explanation, which community will be (dis) advantaged? If it is unbiased, write your explanation.

Question 2: What could be an ethical use of Emotional recognition technology? Explain how that use is ethical?